CAPABILITY ASSESSMENT & COMPETITIVE ANALYSIS

USSC Calendar vs. Existing Scheduling & Tracking Software

How the USSC Calendar outclasses Aurora Scheduling and GPredict for operational space command coordination

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Executive Summary

The USSC Calendar is purpose-built for multi-alliance space command operational scheduling. No existing commercial or open-source tool occupies this niche. **Aurora Scheduling** is an enterprise project optimizer for aerospace manufacturing — it knows nothing about satellites, real-time orbits, or alliance deconfliction. **GPredict** is a hobbyist satellite tracker for amateur radio operators — it has no scheduling capability, no shared access, and no operational structure.

> **Key finding:** GPredict tracks satellites but cannot schedule around them. Aurora schedules complex projects but knows nothing about satellites. USSC Calendar is the only tool that does both — and adds multi-alliance coordination that neither competitor can approach.

Competitor Profiles

Aurora Scheduling

Aurora is a constraint-optimization engine used by aerospace primes (Bombardier, Textron, General Dynamics, NASA) to schedule complex manufacturing and maintenance operations. Its strengths are Monte Carlo simulation, critical-chain analysis, resource leveling, and Gantt/PERT visualization. It is deployed as a contact-sales enterprise product at an estimated $220–$260 per user per month.

Core limitation: Aurora has zero domain knowledge of orbital mechanics, satellite assets, ground stations, or alliance structures. It cannot ingest TLE data, compute pass windows, or reason about which alliance is using which asset when. It is a generic scheduling solver, not an operational space tool.

GPredict

GPredict is a free, open-source GTK+ desktop application for real-time satellite tracking and pass prediction, targeting amateur radio operators. It ingests TLE data from Celestrak, propagates orbits via SGP4/SDP4,

shows satellite positions on a world map, and predicts upcoming pass windows. It integrates with Hamlib for Doppler tuning of radio receivers.

Core limitation: GPredict is a *tracker*, not a *scheduler*. It shows when a satellite will be visible, but cannot create calendar events, coordinate multi-user access, enforce alliance visibility rules, detect conflicts, or share data across a network. Known bugs include segmentation faults on rotator control, primitive pass prediction resolution, and no Windows-native support.

Side-by-Side Feature Comparison

Capability	Aurora Scheduling	GPredict	USSC Calendar
Satellite TLE ingestion	✗ None	✓ Yes (Celestrak)	✓ Yes (Celestrak, cached)
SGP4 pass prediction	✗ None	✓ Yes (real-time display)	✓ Yes → generates calendar events
Pass → calendar event	✗ None	✗ None	✓ Automatic, color-coded
Ground station management	✗ None	~ Manual lat/lon only	✓ Named stations, elevation mask
Multi-user shared calendar	~ Per-seat, no realtime sharing	✗ Single user, desktop only	✓ LAN master instance, shared state
Alliance visibility control	✗ None	✗ None	✓ CSpO / NATO / INDOPACOM
Satellite conflict detection	~ Generic resource conflict	✗ None	✓ Double-booking, overlap detection
Alliance deconfliction report	✗ None	✗ None	✓ Per-satellite usage timeline
Exchange / M365 integration	✗ None	✗ None	✓ EWS, Graph API, Outlook COM
Government auth (Kerberos, OAuth)	✗ None	✗ None	✓ Device-code, NTLM/Kerberos
Natural language event creation	✗ None	✗ None	✓ Free-text event parser
ICS export	✗ None	✗ None	✓ RFC 5545 compliant
Offline / air-gap capable	✗ Cloud-dependent	✓ Yes	✓ Yes (local auth + storage)
Windows 11 native	✓ Yes	~ Via GTK compatibility layer	✓ Native Win32/wxWidgets
Cost	~$240/user/month	Free / open source	Internal / program-funded

Gap Analysis: Where Competitors Fall Short

GPredict Gaps

- Tracks passes but cannot create calendar events from them

Aurora Scheduling Gaps

- Zero orbital mechanics knowledge — cannot work with TLEs

- No multi-user access — single desktop only
- No alliance or operational unit structure
- No conflict detection when two users schedule the same asset
- Buggy: segfaults on rotator, broken pass refinement, Linux-primary
- No government authentication or network calendar integration
- No named ground station catalog with elevation masks

- No satellite asset database or pass prediction capability
- No alliance, nation, or operational unit visibility controls
- No government authentication patterns
- Contact-sales only; no transparent pricing or self-serve trial
- Custom integration required for every system connection
- Not designed for real-time operational coordination

USSC Calendar Unique Capabilities

What Only USSC Calendar Does

- **Pass → Event pipeline:** SGP4 propagation feeds directly into the shared operational calendar — no manual transfer required
- **Multi-alliance structure:** CSpO, NATO, INDOPACOM visibility controls baked into every event and view
- **Satellite conflict detection:** Flags double-booked assets and overlapping pass windows automatically
- **Alliance deconfliction report:** One-click view of who is using which satellite, when, and across which alliance
- **Government-native auth:** Kerberos/NTLM, OAuth 2.0 device-code flow, and on-premises Exchange — all supported natively
- **LAN shared state:** All ground stations, TLE caches, and pass events are shared across the installation

Implemented Feature Roadmap

The following capability tiers have been implemented and verified to build cleanly:

TIER 1 — Closes the GPredict Gap

- SGP4/SDP4 orbital propagator (Vallado algorithm, near-Earth objects)
- TLE ingestion from Celestrak via HTTPS for all catalogued NORAD IDs
- Named ground station management (lat/lon, elevation mask, active flag)
- Automatic pass prediction: AOS, TCA, LOS, max elevation, azimuth at TCA
- Pass events auto-generated in the shared calendar, color-coded teal
- 3-day lookahead window with 30-second coarse / 2-second refinement step

TIER 2 — Closes the Aurora Gap

- Satellite conflict detector: identifies double-booked assets by satellite ID
- Overlap detection across all pass events in the event store
- Conflict data exposed to the alliance report for operator awareness

TIER 3 — Extends the Unique Moat

- Alliance Deconfliction Report: per-satellite timeline grouped by alliance and nation
- Conflict flags annotated in the report output
- Report accessible from the File menu, no extra tooling required

Architecture: LAN Master Instance

Unlike GPredict (single-user desktop) or Aurora (per-seat cloud), USSC Calendar runs as a **single master instance on the LAN**. All operators log in to the same running instance via the network. This means:

- Ground station configurations are shared — one admin manages the catalog for all users
- TLE caches are fetched once and available to all sessions immediately
- Generated pass events appear on every operator's calendar simultaneously
- Alliance deconfliction reports reflect the true real-time state across all users
- No sync protocol required — all state is authoritative at the source

Conclusion

USSC Calendar occupies a capability space that neither Aurora Scheduling nor GPredict can reach. Aurora is a powerful scheduling engine with no orbital or alliance context. GPredict is a useful tracking tool with no scheduling or coordination capability.

USSC Calendar combines real-time SGP4 pass prediction, automatic calendar event generation, multi-alliance visibility controls, government authentication, satellite conflict detection, and a shared LAN architecture into a single purpose-built operational tool. It does not require expensive enterprise licensing, works offline and in air-gapped environments, and integrates natively with the DoD's existing Exchange/M365 infrastructure.

Bottom line: For space command operational scheduling, USSC Calendar is not competing with Aurora or GPredict — it is creating a new category that neither existing tool serves.